

Summary for 2024-10-01-SensitivityDuality

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1 Summary

This lecture on Linear Programming explored sensitivity analysis and duality theory. The core concept is understanding how changes in a linear program's parameters affect its optimal solution and objective function value.

Primal and Dual Problems: The lecture introduced primal and dual linear programming problems, demonstrating their interconnectedness through strong duality, where the optimal objective function values are equal but opposite in sign.

Complementary Slackness: Complementary slackness conditions were used to verify the optimality of a solution by relating the primal and dual variables.

Sensitivity Analysis: The lecture focused on analyzing the sensitivity of the optimal solution to changes in the right-hand side of the constraints, using the B, N method and matrix manipulations.

Shadow Prices: The optimal dual variables were interpreted as shadow prices, representing the rate of change in the optimal objective function value with respect to changes in the right-hand side constraints.

Range of Optimality: The lecture determined the range of changes to the right-hand side for which the current optimal basis remains optimal. This was achieved by analyzing the impact of changes on both primal and dual feasibility.

Matrix Methods: Matrix notation and operations, including the inverse of the basis matrix, were extensively used to efficiently analyze the sensitivity of the optimal solution and objective function value.